

Module 01 — OOP, SOLID & UML

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Week 2 · Prerequisites: m00 (records, sealed interfaces, switch patterns, lambdas) · Estimated study time: 5 h

```
Run          every          example          without          a          build:
java modules/m01-oop-solid-uml/src/main/java/io/github/aliturgutbozkurt/patterns/
m01/examples/<path>/<Demo>.java
(JDK 27)
```

1 Learning outcomes

By the end of this module you can:

1. **Explain** the four OOP pillars and **judge** the coupling and cohesion of a class.
2. **Identify** a violation of each SOLID principle and **refactor** it without changing behaviour, protected by a characterization test.
3. **Decide** between inheritance and composition, and **implement** a forwarding wrapper.
4. **Inject** dependencies — including time, through `java.time.Clock` — so code can be tested without side effects.
5. **Draw** UML class and sequence diagrams in Mermaid and **read** the diagrams used in the rest of the course.
6. **Describe** the GoF catalog and find where each pattern is taught in this course.

2 Motivation

Every design pattern in this course is justified with the same handful of ideas: *this class has too many reasons to change, this code must be edited for every new case, this subclass surprises its callers, this client depends on methods it never uses, this policy is welded to a detail*. Those five sentences are the SOLID principles. Learn to hear them in code and the patterns stop being recipes to memorise — they become the obvious fix.

Every principle below is taught the same way: a small **before** version with the problem, an **after** version that fixes it, and a test that proves the two behave the same. Each demo prints both versions.

3 OOP pillars, coupling and cohesion

3.1 The four pillars

Pillar	Meaning	Example in this module
Encapsulation	Hide state; let the object guard its own invariants	<code>CheckingAccount</code> never lets its balance go negative
Abstraction	Expose <i>what</i> an object does, not <i>how</i>	<code>MessageSender</code> says "send a message", not "open an SMTP socket"
Inheritance	A subtype reuses and extends a supertype	<code>Square</code> extends <code>Rectangle</code> — and how it goes wrong
Polymorphism	One call, many behaviours chosen at run time	<code>PrintQueue</code> prints on any <code>Printer</code> , even a lambda

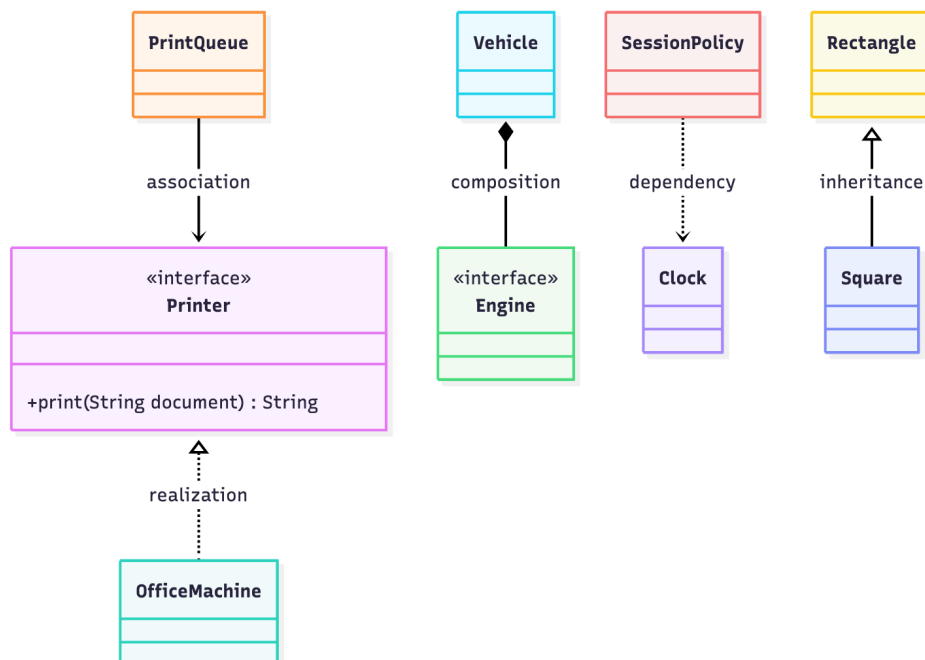
3.2 Coupling and cohesion

Cohesion is how closely the parts of one class belong together. **Coupling** is how much one class knows about another. Good designs have *high cohesion* and *low (loose) coupling*: each class does one thing well, and a change inside it does not ripple through the system. Every SOLID principle is a way to raise cohesion or lower coupling.

4 UML in Mermaid

This course draws diagrams as text with [Mermaid](#), so they live next to the code, render on GitHub and are pre-rendered in the PDFs.

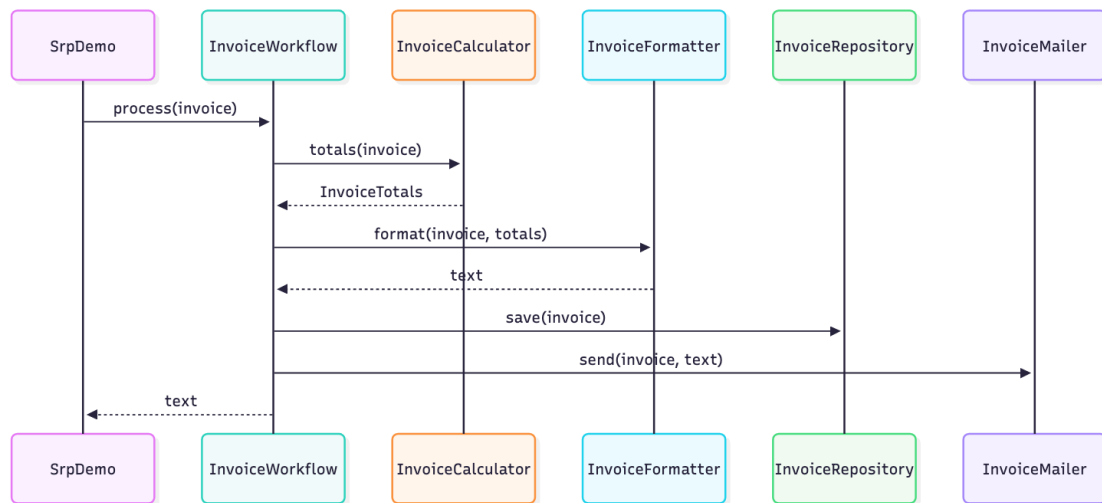
4.1 Class diagrams



Arrow	Mermaid	Reads as
Inheritance	A < -- B	B is an A (extends)
Realization	A < .. B	B implements interface A
Association	A --> B	A holds a reference to B
Aggregation	A o-- B	A has B, but B can live without A
Composition	A *- B	A owns B; B's lifetime is A's
Dependency	A ..> B	A uses B briefly (parameter, local variable)

4.2 Sequence diagrams

A sequence diagram shows *who calls whom, in which order*. Time runs downwards:



5 Single Responsibility Principle (SRP)

5.1 Problem

`InvoiceService` computes totals, formats the invoice, stores it and e-mails it. A new tax rule, a new layout, a database and a new mail provider are four unrelated reasons to open the same class — and four ways to break the others.

```
// file: examples/srp/before/InvoiceService.java
public final class InvoiceService {
    // ...
    /** Calculates, formats, stores and "sends" the invoice; returns its text. */
    public String process(String number, String customer, List<Line> lines) {
        // ...
        // 1) calculation
        BigDecimal subtotal = BigDecimal.ZERO;
        // ...
        // 2) formatting
        var text = new StringBuilder();
        // ...
        // 3) storage
        sentInvoices.put(number, result);
        // 4) delivery
    }
}
```

```

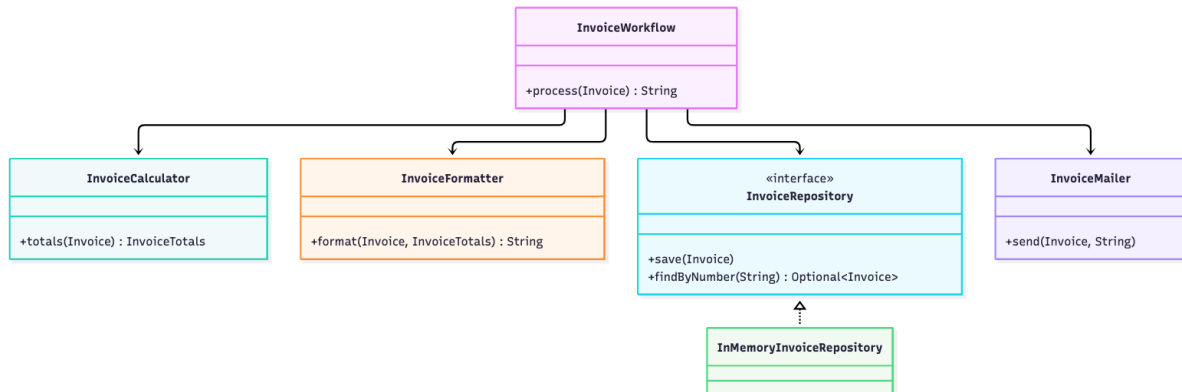
        System.out.println("Emailing invoice " + number + " to " + customer);
        return result;
    }

```

5.2 Principle

A class should have **one reason to change** — it should answer to one actor (the tax office, the designer, the DBA, ...).

5.3 Structure



5.4 After

Each collaborator now has one job. The workflow only fixes the *order* of the steps:

```

// file: examples/srp/after/InvoiceWorkflow.java
public final class InvoiceWorkflow {
    // ...
    /** Calculates, formats, stores and sends the invoice; returns its text. */
    public String process(Invoice invoice) {
        String text = formatter.format(invoice, calculator.totals(invoice));
        repository.save(invoice);
        mailer.send(invoice, text);
        return text;
    }
}

```

The money rules live in one small class that a tax change — and only a tax change — touches:

```

// file: examples/srp/after/InvoiceCalculator.java
public final class InvoiceCalculator {

    /** 20 % VAT (Turkish KDV). */
    public static final BigDecimal VAT_RATE = new BigDecimal("0.20");

    public InvoiceTotals totals(Invoice invoice) {
        BigDecimal subtotal = invoice.lines().stream()
            .map(InvoiceLine::amount)
            .reduce(BigDecimal.ZERO, BigDecimal::add)
            .setScale(2, RoundingMode.HALF_EVEN);
        BigDecimal vat = subtotal.multiply(VAT_RATE).setScale(2,
            RoundingMode.HALF_EVEN);
    }
}

```

```

        return new InvoiceTotals(subtotal, vat, subtotal.add(vat));
    }
}

```

How do we know the refactoring did not change anything? `InvoiceTest.refactoredWorkflowBehavesExactlyLikeTheGodClass` runs both versions and compares their output. A test that pins down *existing* behaviour before a refactoring is called a **characterization test**. `SrpDemo` prints both versions:

```

INVOICE INV-001
Customer: Ada Lovelace
  2 x Keyboard      49.90      99.80
  1 x Mouse         19.99      19.99
  1 x Monitor       229.00     229.00
Subtotal:      348.79
VAT 20%:       69.76
Total:         418.55
same text? true

```

5.5 A second example: the gradebook

`GradeBook.report` parses CSV lines, computes averages, maps them to letter grades and prints a table — in one method. The refactoring separates the **input format** (`ScoreParser`), the **grading policy** (`GradingScale`) and the **layout** (`GradeReport`). When the faculty moves a grade boundary, only this class changes:

```

// file: examples/srp/gradebook/after/GradingScale.java
/** Letter grade for an average in {@code [0, 100]}. */
public String letterFor(BigDecimal average) {
    if (average.signum() < 0 || average.compareTo(BigDecimal.valueOf(100)) > 0) {
        throw new IllegalArgumentException("average out of range 0-100: " +
average);
    }
    return BOUNDARIES.stream()
        .filter(boundary -> average.compareTo(boundary.minimum()) >= 0)
        .map(Boundary::letter)
        .findFirst()
        .orElse("FF");
}

```

```

// file: examples/srp/GradeBookDemo.java
String after = new GradeReport(new GradingScale()).render(new
ScoreParser().parse(CSV));

```

5.6 Real-world usage

`java.time` keeps values (`LocalDate`), formatting (`DateTimeFormatter`) and time sources (`Clock`) in separate types. JDBC separates connecting (`DataSource`), running a statement (`PreparedStatement`) and reading rows (`ResultSet`).

5.7 Pitfalls and when NOT to apply it

- "One reason to change" is not "one method". Splitting every method into its own class lowers cohesion again.
- Split when two responsibilities actually change at different times or for different people — not in advance.

6 Open/Closed Principle (OCP)

6.1 Problem

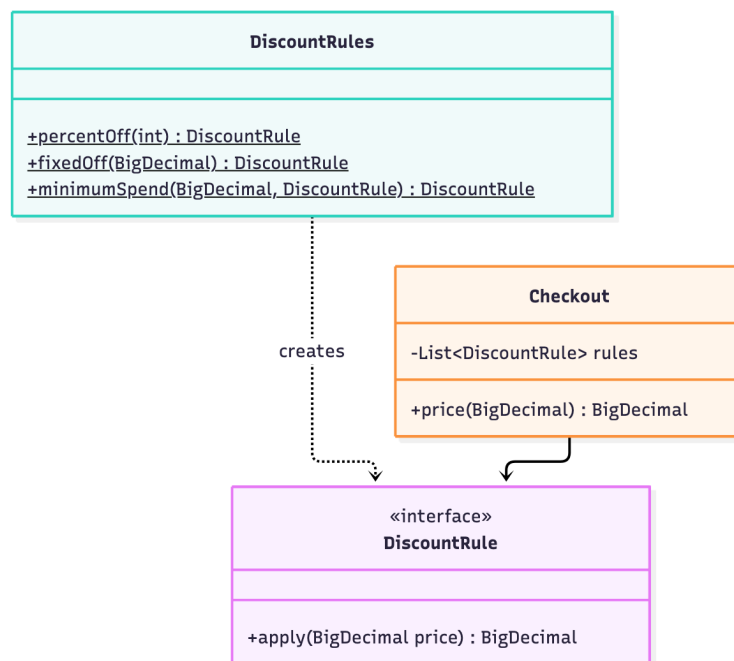
Every new kind of discount means opening `PriceCalculator` and adding another branch — and re-testing every old one:

```
// file: examples/ocp/before/PriceCalculator.java
BigDecimal percentOff;
if (customerType.equals("REGULAR")) {
    percentOff = BigDecimal.ZERO;
} else if (customerType.equals("STUDENT")) {
    percentOff = BigDecimal.valueOf(10);
} else if (customerType.equals("VIP")) {
    percentOff = BigDecimal.valueOf(15);
} else {
    throw new IllegalArgumentException("unknown customer type: " +
customerType);
}
```

6.2 Principle

Software entities should be **open for extension, closed for modification**: new behaviour is added as new code, not by editing code that already works.

6.3 Structure



6.4 After

The extension point is a functional interface, so a new rule can be a lambda:

```
// file: examples/ocp/after/DiscountRule.java
@FunctionalInterface
public interface DiscountRule {
```

```

    /** Returns the price after this discount. */
    BigDecimal apply(BigDecimal price);
}

```

`Checkout` applies whatever rules it is given and never needs to know which ones exist:

```

// file: examples/ocp/after/Checkout.java
public BigDecimal price(BigDecimal amount) {
    if (amount.signum() < 0) {
        throw new IllegalArgumentException("amount must not be negative: " +
amount);
    }
    BigDecimal price = amount;
    for (DiscountRule rule : rules) {
        price = rule.apply(price);
    }
    return price.max(BigDecimal.ZERO).setScale(2, RoundingMode.HALF_EVEN);
}

```

A Black Friday rule arrives as two lines in `OcpDemo` — `Checkout` is not touched:

```

// file: examples/ocp/OcpDemo.java
DiscountRule roundDownToWholeLira = price -> price.setScale(0,
RoundingMode.FLOOR);
var blackFriday = new Checkout(List.of(percentOff(15), roundDownToWholeLira));

```

6.5 A second example: sorting courses

`CourseSorter` switches on a sort-key string. The JDK already solved this problem: `Comparator` is an extension point, and `comparing`, `thenComparing` and `reversed` compose new orderings from old ones. `CourseCatalog` simply accepts any `Comparator`:

```

// file: examples/ocp/CourseSortDemo.java
Comparator<Course> byDepartmentThenCredits = comparing(Course::department)
    .thenComparing(comparingInt(Course::credits).reversed())
    .thenComparing(Course::code);

```

6.6 Sidebar: sealed types are deliberately closed

A sealed interface with an exhaustive `switch` is the *opposite* trade-off. Adding a new `Engine` forces you to edit every `switch` over it, and the compiler lists each one:

```

// file: examples/composition/vehicles/after/Vehicle.java
double range = switch (engine) {
    case Engine.Petrol(double tank, double per100) -> tank * 100 / per100;
    case Engine.Diesel(double tank, double per100) -> tank * 100 / per100;
    case Engine.Electric(double battery, double per100) -> battery * 100 /
per100;
};

```

This is the *expression problem*. An open interface makes new **types** cheap and new **operations** expensive; a sealed type with `switch` makes new **operations** cheap and new **types** expensive. Choose based on which one you expect to add more often. Module `m08` (Visitor vs. pattern matching) returns to this.

6.7 Real-world usage

`Comparator`, `Collector`, `java.util.function`, and `ServiceLoader` (m02) are all extension points: the JDK is closed for modification, yet you extend its behaviour every day.

6.8 Pitfalls and when NOT to apply it

- Do not add extension points "just in case". The first variation can be an `if`; extract a rule interface when the second or third appears.
- Rules applied in sequence interact: the tests `appliesRulesInOrder` show that 10 % then -20 is not -20 then 10 %.

7 Liskov Substitution Principle (LSP)

7.1 Problem

Geometrically a square *is a* rectangle. As a mutable subtype it is not: to keep its sides equal, `Square` must change both sides in each setter.

```
// file: examples/lsp/before/Square.java
public final class Square extends Rectangle {

    public Square(double side) {
        // Flexible constructor body (JEP 513): validate with Square's own message
        before super(...) runs.
        if (!(side > 0)) {
            throw new IllegalArgumentException("side must be positive: " + side);
        }
        super(side, side);
    }

    @Override
    public void setWidth(double width) {
        super.setWidth(width);
        super.setHeight(width);
    }
    // ...
}
```

A client written against `Rectangle` reasonably expects 20 — and gets 16 for a `Square`:

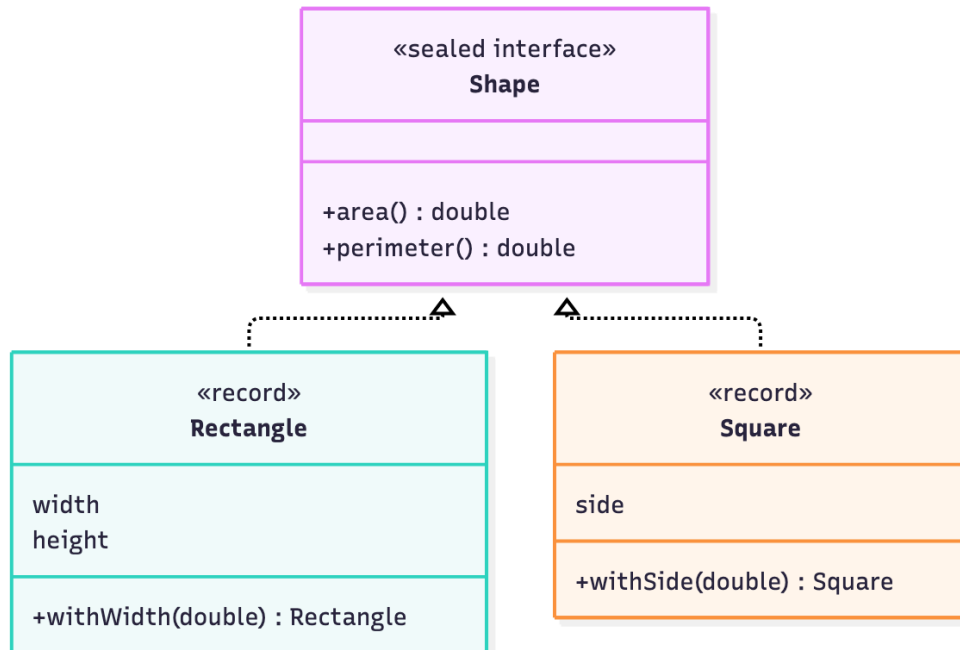
```
// file: examples/lsp/before/RectangleClient.java
/** Resizes to 5 × 4 and returns the area — the client reasonably expects 20. */
public static double resizeTo5By4(Rectangle rectangle) {
    rectangle.setWidth(5);
    rectangle.setHeight(4);
    return rectangle.area();
}
```

Note the constructor: since Java 25 (JEP 513) statements may run **before** `super(...)`, so `Square` validates its side with its own error message before the `Rectangle` constructor runs.

7.2 Principle

Objects of a subtype must be usable **wherever the supertype is expected**, without the client noticing. A subtype may not strengthen preconditions, weaken postconditions or break the supertype's invariants.

7.3 Structure



7.4 After

Immutable records have no setters, so there is no setter contract to break. `Square` and `Rectangle` become siblings:

```
// file: examples/lsp/after/Rectangle.java
public record Rectangle(double width, double height) implements Shape {
    // ...
    public Rectangle withWidth(double newWidth) {
        return new Rectangle(newWidth, height);
    }
}
```

7.5 A second example: bank accounts

`FixedDepositAccount` extends `Account` but refuses every withdrawal — it **strengthens the precondition** of `withdraw` from "amount ≤ balance" to "never":

```
// file: examples/lsp/accounts/before/FixedDepositAccount.java
@Override
public void withdraw(BigDecimal amount) {
    throw new UnsupportedOperationException("no withdrawals from a fixed deposit");
}
```

The fix models the capability instead of refusing it. `Account` only promises what every account can do; `Withdrawable` is a separate role, and `BillPayer` asks for exactly that role:

```
// file: examples/lsp/accounts/after/BillPayer.java
/** Pays a bill and returns the remaining balance. */
public static BigDecimal pay(Withdrawable source, BigDecimal bill) {
    source.withdraw(bill);
    return source.balance();
}
```

Passing a `FixedDepositAccount` to `pay` is now a compile error, not a run-time surprise. Splitting an interface by what clients need is the next principle.

7.6 Real-world usage

`List.of(...)` returns a `List` whose `add` throws `UnsupportedOperationException`. The JDK documents such methods as "optional operations" — a pragmatic trade-off, but exactly the surprise LSP warns about.

7.7 Pitfalls and when NOT to apply it

- "Is-a" in the real world is not enough; ask "is it **substitutable** for" the supertype's behaviour.
- A test suite written against the supertype (like the contract tests in the assignments) is the best LSP check: every implementation must pass it.

8 Interface Segregation Principle (ISP)

8.1 Problem

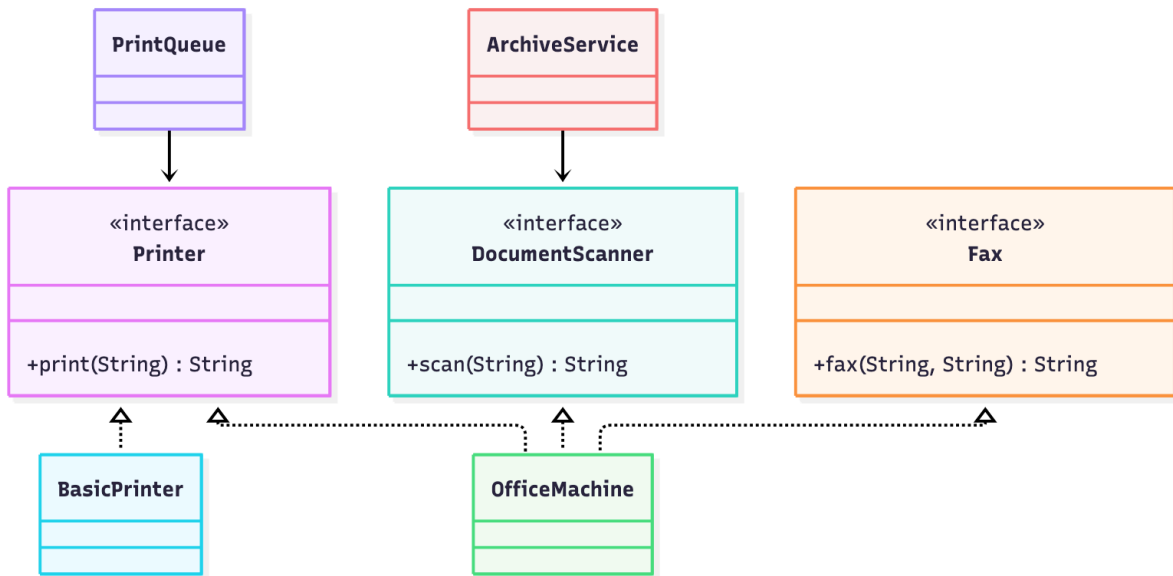
`MultiFunctionDevice` forces every device to print, scan and fax. A plain printer can only throw:

```
// file: examples/isp/before/BasicPrinter.java
@Override
public String scan(String page) {
    throw new UnsupportedOperationException("BasicPrinter cannot scan");
}
```

8.2 Principle

Clients should not be forced to depend on methods they do not use. Prefer several small **role interfaces** to one large one.

8.3 Structure



8.4 After

Each role is one small interface — small enough to be a lambda. One class may still play several roles:

```
// file: examples/isp/after/Printer.java
@FunctionalInterface
public interface Printer {

    String print(String document);
}
```

```
// file: examples/isp/after/OfficeMachine.java
public final class OfficeMachine implements Printer, DocumentScanner, Fax {
```

A client names the smallest role it needs:

```
// file: examples/isp/after/PrintQueue.java
public PrintQueue(Printer printer) {
    this.printer = Objects.requireNonNull(printer, "printer");
}
```

8.5 A second example: a product store

A read-only view of a five-method `ProductStore` must "implement" `save`, `delete` and `importAll` by throwing. After the split, `CatalogPage` depends on `ProductReader` only, so write methods are not even visible to it, and `InMemoryProducts` implements both roles. A default method keeps the writer role convenient:

```
// file: examples/isp/store/after/ProductWriter.java
/** Saves every product in order. */
default void importAll(List<Product> products) {
    products.forEach(this::save);
}
```

8.6 Real-world usage

The JDK is full of role interfaces: `Readable`, `Appendable`, `AutoCloseable`, `Comparable`, `Iterable`. A `StringBuilder` and a `Writer` are both `Appendable`, so code that only appends works with either.

8.7 Pitfalls and when NOT to apply it

- Do not split below what clients use together; one method per interface everywhere makes wiring painful.
- Segregate by **client need**, not by implementation detail.

9 Dependency Inversion Principle (DIP)

9.1 Problem

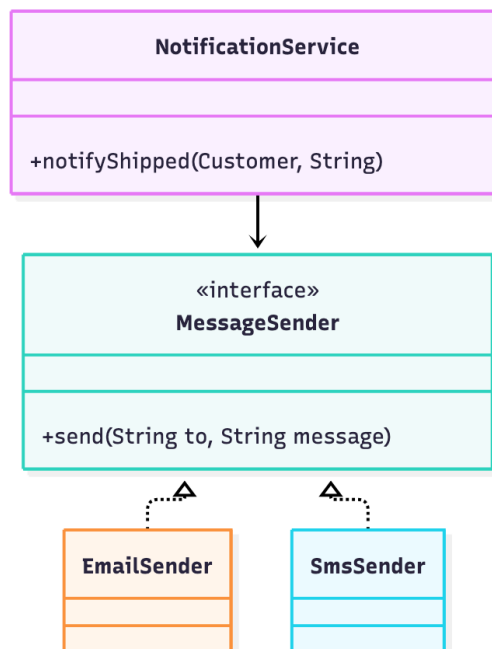
The business rule "tell the customer the order shipped" creates a concrete e-mail sender itself. It cannot send SMS, and it cannot be tested without really "sending":

```
// file: examples/dip/before/NotificationService.java
public void notifyShipped(Customer customer, String orderId) {
    var sender = new EmailSender(); // hard-wired dependency on a concrete class
    sender.send(customer.email(), "Order " + orderId + " has shipped, " +
customer.name() + ".");
}
```

9.2 Principle

High-level policy should not depend on low-level details; **both depend on abstractions** — and the abstraction is owned by the high-level side.

9.3 Structure



The arrows from `EmailSender` and `SmsSender` now point *up* to an interface that lives next to `NotificationService` — that is the "inversion".

9.4 After

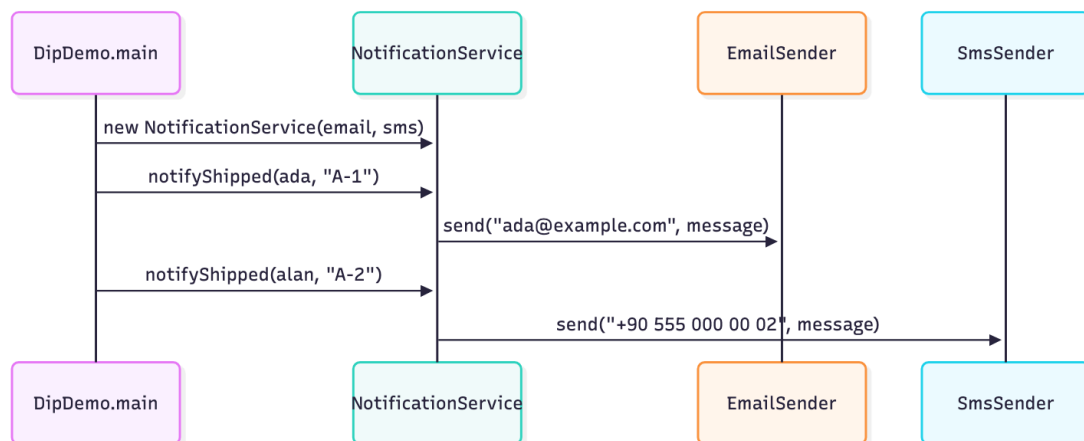
```
// file: examples/dip/after/NotificationService.java
public NotificationService(MessageSender email, MessageSender sms) {
    this.email = Objects.requireNonNull(email, "email");
    this.sms = Objects.requireNonNull(sms, "sms");
}

public void notifyShipped(Customer customer, String orderId) {
    String message = "Order " + orderId + " has shipped, " + customer.name() + ".";
    switch (customer.preferred()) {
        case EMAIL -> email.send(customer.email(), message);
        case SMS -> sms.send(customer.phone(), message);
    }
}
```

Someone still has to call `new`. That place is the **composition root** — here, `main`:

```
// file: examples/dip/DipDemo.java
var service = new NotificationService(new EmailSender(), new SmsSender());
```

The call sequence at run time:



In the tests, a recording `MessageSender` (a **test double**) replaces the real senders — no console capture needed.

9.5 A second example: time as a dependency

`Instant.now()` inside business logic is a hidden dependency on the system clock. The JDK's own abstraction is `java.time.Clock`; inject it and tests can pin time to any instant:

```
// file: examples/dip/clock/SessionPolicy.java
/** Expired from the instant {@code startedAt + timeout} onwards. */
public boolean isExpired(Session session) {
    return !clock.instant().isBefore(session.startedAt().plus(timeout));
}
```

```
session for ada started at 2026-09-29T09:00:00Z, timeout PT30M
at 2026-09-29T09:29:59Z -> active
at 2026-09-29T09:30:00Z -> expired
```

9.6 Real-world usage

JDBC code depends on `javax.sql.DataSource`, not on a specific driver; logging code depends on `System.Logger`, not on a logging library. Dependency Injection frameworks automate the composition root — module m11 builds one by hand.

9.7 Pitfalls and when NOT to apply it

- Not every `new` is a problem: values (`BigDecimal`, records) and stable JDK types are fine to create directly.
- One interface per class "for testability" with a single implementation forever is ceremony; invert where a real variation or a slow/side-effecting dependency exists.

10 Composition over inheritance

10.1 Problem

`CountingSet` extends `HashSet` wants to count how many elements were ever added:

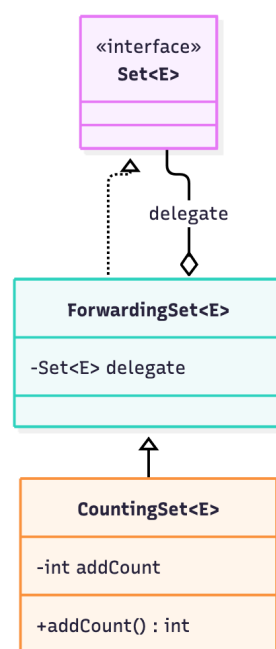
```
// file: examples/composition/before/CountingSet.java
@Override
public boolean addAll(Collection<? extends E> elements) {
    addCount += elements.size();
    return super.addAll(elements); // calls this.add(...) for each element ->
    counted again
}
```

`HashSet.addAll` happens to call `add` for each element — our overridden `add`. Three elements are counted six times. The subclass depends on an implementation detail of its parent: the **fragile base class** problem.

10.2 Principle

Prefer **has-a** (composition) to **is-a** (inheritance) when you want to reuse behaviour. Inherit only for a true subtype relationship that you control.

10.3 Structure



10.4 After

`ForwardingSet` implements `Set` by forwarding every call to a wrapped set. `CountingSet` extends the forwarder — a class we control — and wraps any `Set`:

```
// file: examples/composition/after/ForwardingSet.java
public class ForwardingSet<E> implements Set<E> {

    private final Set<E> delegate;

    public ForwardingSet(Set<E> delegate) {
        this.delegate = Objects.requireNonNull(delegate, "delegate");
    }
    // ...
    @Override public boolean add(E e) { return delegate.add(e); }
    // ...
    @Override public boolean addAll(Collection<? extends E> c) { return
delegate.addAll(c); }
```

The wrapped set's `addAll` calls *its own* `add`, not ours, so each element is counted once. The same shape returns as **Decorator** in m04.

10.5 A second example: vehicles

With inheritance, every engine × gearbox combination needs its own subclass — $2 \times 2 = 4$ classes, and an electric engine would add 2 more. With composition, a vehicle *has* an engine and a gearbox, and each dimension varies on its own:

```
// file: examples/composition/vehicles/after/Vehicle.java
public record Vehicle(Engine engine, Gearbox gearbox) {
```

Adding the electric engine took one new record; `VehicleDemo` prints all six combinations.

10.6 Real-world usage

`java.util.Stack` extends `Vector` and `java.util.Properties` extends `Hashtable` are the JDK's own regrets: both inherit methods that break their intended use. `Collections.unmodifiableSet(...)` and `Collections.synchronizedList(...)`, by contrast, are wrappers — composition.

10.7 Pitfalls and when NOT to apply it

- A forwarding class is boilerplate; it pays off when the wrapped type is not yours or has many implementations.
- Inheritance is still right for a genuine, stable "is-a" inside your own code, especially with `sealed` hierarchies where you control every subtype.

11 The GoF catalog

In 1994 Gamma, Helm, Johnson and Vlissides ("Gang of Four") catalogued 23 patterns in three families. This course teaches all of them, each in modern Java:

Family	Pattern	Module
Creational	Singleton, Factory Method, Abstract Factory	m02
Creational	Builder, Prototype	m03
Structural	Adapter, Decorator, Proxy	m04
Structural	Composite, Bridge, Facade, Flyweight	m05
Behavioral	Strategy, Template Method, Command, Iterator	m06
Behavioral	Observer, Mediator, Chain of Responsibility, Memento	m07
Behavioral	State, Visitor, Interpreter	m08

Several of them you have already met in this module without the name: `DiscountRule` is a **Strategy**, `ForwardingSet` is the skeleton of a **Decorator**, and `InvoiceWorkflow` is close to a **Facade**.

12 Summary

Principle	Smell	Fix	Java 27 shortcut
SRP	A class changes for several unrelated reasons	Split by reason to change; keep a coordinator	Records for the data
OCP	A new case edits a working class	An extension point (interface)	Functional interface + lambda
LSP	A subtype surprises its callers	Model capabilities; immutable values	Sealed records, no setters
ISP	Implementers throw <code>UnsupportedOperationException</code>	Small role interfaces	<code>@FunctionalInterface</code> roles
DIP	Policy calls <code>new</code> on a detail	Inject an abstraction owned by the policy	<code>java.time.Clock</code> , lambdas as doubles
Composition	Subclass depends on parent internals	Wrap and forward	Records composed of sealed parts

13 Quiz

- `InvoiceFormatter` and `InvoiceCalculator` are both used by `InvoiceWorkflow`. Which actor would make each of them change?
- What is a characterization test, and why write one *before* refactoring?
- Why does a sealed interface with an exhaustive `switch` not violate OCP in a harmful way?
- `Square` extends `Rectangle` breaks LSP. Would it still break LSP if both classes were immutable? Why?
- Which LSP rule does `FixedDepositAccount.withdraw` break: preconditions, postconditions or invariants?
- Give a JDK example of an ISP-friendly role interface and of an "optional operation".
- In DIP, who owns the `MessageSender` interface, and where are the concrete senders created?
- Why does `CountingSet` extends `HashSet` count `addAll` twice, and why can the forwarding version not have this bug?
- Draw the Mermaid arrow for: "`Vehicle` owns an `Engine`" and "`SessionPolicy` uses a `Clock` passed in".

Answers

1. The tax office (VAT rules) changes the calculator; whoever designs the invoice layout changes the formatter.
2. A test that records what the code does *today*. It makes a refactoring safe: if the output changes, the test fails.
3. It is a deliberate choice: new *operations* are easy and the compiler lists every place a new *type* must be handled. OCP is about the axis you expect to change.
4. No. Without setters there is no "set width only" contract to violate; `getWidth` returns a new value.
5. It strengthens the precondition: callers that satisfy `Account`'s precondition (`amount ≤ balance`) still fail.
6. `Appendable` / `AutoCloseable` / `Comparable`; `List.of(...).add(...)` throws for an optional operation.
7. The high-level side owns it (it sits next to `NotificationService`); the senders are created in `main`, the composition root.
8. `HashSet.addAll` calls the overridden `add`. The forwarding version calls `delegate.addAll`, which calls the delegate's own `add`, never ours.
9. `Vehicle` *-- `Engine` (composition) and `SessionPolicy` ..> `Clock` (dependency; --> if it is kept in a field, as here).

14 Assignments

- [01 – Sales report: SRP + OCP refactoring](#) ★★☆
- [02 – Library loans: DIP + LSP](#) ★★☆

15 Further reading

- Robert C. Martin, "The Principles of OOD" and *Clean Architecture* (2017), part III — the source of the SOLID names.
- Barbara Liskov and Jeannette Wing, "A Behavioral Notion of Subtyping" (1994).
- Gamma, Helm, Johnson, Vlissides, *Design Patterns* (1994), chapter 1 — "Favor object composition over class inheritance".
- JEP 513 — [Flexible Constructor Bodies](#)
- Mermaid — [class diagrams](#) · [sequence diagrams](#)
- Terms in Turkish: [docs/glossary.md](#)